

## **What Linux Distribution should I choose?**

Choosing a Linux distribution is a personal thing. It greatly depends on what you want to do with it.

This is a short collection of some of the more popular distributions out there. For more in-depth information on the differences between each 'distro', we would recommend visiting [distrowatch.com](http://distrowatch.com)

Here is an example of just some of the more popular Linux vendors today:

### **Ubuntu and Linux Mint**

Suitable for: Beginner to Advanced/Server

Ubuntu is one of the most popular Linux Distributions today. It is built on a Debian core, but has a more regular release cycle. It is arguably more polished than Debian, is easier to use and has major financial backing. Ubuntu is a completely free distro, therefore copyrighted materials such as DVD & MP3 playing 'codecs' do not come as standard with Ubuntu, you must download and install it separately, but can be done easily. Due to Ubuntu's prevalence and ease of use, I have based the later chapters on installing and using Ubuntu.

Canonical, the company that make and back Ubuntu, also offer enterprise level support for Ubuntu. There are three main versions, Desktop, Server and Core. However, there are lots of other 'spins' available such as 'Edubuntu', made specifically with primary and secondary education in mind and Kubuntu (KDE version), Lubuntu (a lightweight version). In addition to the official spins, there are over 40 third-party versions in circulation today!

Ubuntu is released twice a year (in April and October). Every two years a 'Long Term Support' release is provided in the April release. These offer five years support at no extra cost. This includes security patches and bugfixes. Normal 'intermediary' releases offer 9 month support, but upgrading from one release to the next is an easy process.

If you don't like the look and feel of the Ubuntu desktop, Linux Mint is based on Ubuntu, it is made for beginners and still offers a GNOME, KDE and 'Cinnamon' version. Linux Mint works very similarly to Ubuntu.

### **Fedora/Red Hat/CentOS**

Suitable for: Beginner to Advanced/Server

The company Red Hat was founded around 1993. They have become arguably the most commercially successful Linux based company in the world and are now owned by IBM.

Red Hat Linux had nine major (free) releases until Red Hat decided to take a more corporate approach to Linux in 2003. They then created Red Hat Enterprise Linux (RHEL). This product is used by businesses around the world and is a fully supported, commercial Linux distribution. Most users of RHEL use it as a server operating system, rather than a desktop one.

CentOS is the free version of RHEL and it is said to be 'binary compatible' (meaning it has the exact same software). A lot of businesses use CentOS if they don't need the commercial support or backing from Red Hat.

When Red Hat moved to a more corporate model in 2003, it also released Fedora. Fedora is Red Hat's cutting edge, completely free desktop Linux distribution. It uses the GNOME desktop by

default, however just like Ubuntu, there are many 'spins' on this and a dizzying number of different Fedora versions exist to be downloaded. As Fedora is bleeding-edge, it can also suffer from less stability than their enterprise-grade counterparts as well as other distributions like Debian or Ubuntu LTS releases. I also find that all of the Red Hat system installers are counter-intuitive and needs a good user experience person to give it a once over!

**“Fedora has a reputation for focusing on innovation, integrating new technologies early on and working closely with upstream Linux communities.”**

### **Elementary OS / Solus / Zorin**

Suitable for: Beginner to Intermediate

elementary OS is an Ubuntu-based desktop distribution. I really like it because it's incredibly intuitive for a new user coming from another system (especially from macOS!). Some of its more interesting features include a custom desktop environment called Pantheon which takes cues from the look and feel of macOS. It has many custom apps including Photos, Music, Videos, Calendar, Terminal, Files, and more. It has swapped out some of the standard apps. For example, Firefox has been swapped out for the Epiphany web browser and the email app has been changed for a customised email app based on Geary, much of this is done for aesthetic reasons; if you want Firefox or Thunderbird, for example, you can simply install them through the Software Centre.

If you like distros which have a common look and feel to Windows or Mac, or perhaps just something with a really polished, but familiar look and feel, also be sure to check out Zorin, Solus, and Deepin. These distributions are highly customised to give experiences unlike other distributions. For example, Solus has its own window manager called Budgie, and it was built completely from scratch, not derived from Ubuntu or Fedora.

### **OpenSuSE**

Suitable for: Beginner to Advanced/Developer

SuSE was once an independent German Linux distribution, which later was purchased by Novell, who were acquired by Micro Focus. They have since been acquired and sold a number of times and have a net worth of over \$USD 2.5 billion.

Like RedHat, SuSE split its distribution models into a fully enterprise version. SuSE is an excellent all-rounder which is geared up for the Enterprise which includes support and has corporate partnerships with companies such as SAP. The corporate model isn't the only thing SuSE shares with Red Hat, SuSE was originally based on Red Hat Linux and therefore shares the same RPM based package management system, but over the years it has changed itself enough to make it clearly a distribution in its own right. It's YAST configuration management system makes configuration of services a breeze.

The completely free version of SuSE is called OpenSuSE. OpenSuSE has a lot of interesting features including a 'bleeding-edge' software package system called Tumbleweed. There are a lot of Developer tools baked in too, openQA is built for automated software testing, while Kiwi creates Linux images for deployment on real hardware. OpenSUSE uses the KDE desktop by default.

If you are a developer, definitely give OpenSuSE a try, however I have found recent versions to be buggy and the installation process a bit harder than it should be.

## **Arch Linux, Manjaro and Slackware**

Suitable for: Advanced to Server Users

Arch Linux, unlike most, is not derived from a parent Linux distribution like Red Hat or Debian. It stands alone and is revered by geeks for being a blazing fast distro because it is based on a simple (yet solid) base. Everything else can be added through its pacman packaging system.

Manjaro is a separate distribution that has an Arch based core. It claims to be a user-friendly and desktop based distribution. Both Arch and Manjaro run on a rolling-release mechanism, meaning as long as you keep the system updated, you are always running the very latest version of the distro; there is no need to download the newest version from the website every time a new release comes out.

It's also worthy to mention Slackware here too. Slackware was probably the first real Linux distribution, starting back in 1993! Similarly to Arch and Manjaro, it uses .tar.gz packages rather than more popular APT or YUM systems. If you fall into the more advanced camp, but don't like the sound of compiling everything, perhaps Arch or Mandriva is for you, as it still offers similar levels of customisation as Slackware.

If you are starting out with Linux, then I believe that Arch, Manjaro and Slackware are probably not the best choice. Once you get into Linux, you may want something that is at the bleeding edge and is very fast. You might find yourself a fan of tweaking 'all the things' like a car enthusiast might.

## **Debian**

Suitable for: Intermediate to Advanced Users

Debian Logo

Debian is the granddaddy Linux distro of so many offshoots, including Ubuntu. It was originally released in September 1993. The early distinctions that Debian had over Red Hat based systems was that it had a massive (~50,000 software package library) and secondly that it has an auto-dependency software packaging system called apt. This meant that rather than having to download loads of application packages individually, you could simply tell Debian what app you wanted and it would download the rest for you automatically. It took Red Hat etc a long time to get up to pace with this! . Traditionally known for being further behind than some other distros in terms of having the most up to date packages, it offsets by this by having good stability as the main packages are well tested.

### **Tux tip!**

Note: If a distribution is at release 10 (ie: Slackware), but another distribution is only at 4.1 (ie: Debian), this does not indicate that Debian is an old version of Linux. The release numbers are only an indicator of how many releases that particular vendor has made. For example it is quite likely that Debian 4.1 and Slackware 10 share the same major kernel version and many similar software titles.

Check out The DistroWatch Linux distribution popularity rank (Page Hit Ranking) for a good idea on what's hot in the world of Linux Distributions right now, it's updated daily!